



Course Syllabus

First Semester, Academic Year 2025-2026

Course No. & Descriptive Title: **CSPC - 112 Software Engineering 2**
Name of Faculty Member/s: **Ezekiel O. Bacungan**

Schedule and Room Assignment
4A 08:00-09:00 MF - LR114, 11:00-01:30 W - CLR205
4B 09:00-10:00 MF - LR114, 08:00-09:30 TTh - CLR204
4C 01:00-02:00 MF - LR115, 12:30-02:00 TTh - CLR205

Raymund E. Dilan

4D 08:00-09:30 MF - 205, 02:00-03:00 MF - 113
4E 08:00-10:00 W - 301, 08:00-9:00 TTh - 205
4F 09:30- 11:00 MF - 205, 08:00-10:00 W - 301
4G 10:00-12:00 W - 101, 11:00-12:30 TTh - ilab

Part I:

Philosophy	Total human development with appropriate competencies.
Vision	By 2030, DMMSU IS A globally ranked university.
Mission	Excellence in instruction, research and extension.
Goal	To lead in transforming human resources into productive, self-reliant citizens, and responsible leaders.
Graduate Attributes	DMMSU aims to produce highly competent, specialized, and globally competitive professionals. This is operationalized into graduate attributes that should be attained by the time students will graduate from the University. 1. Professionally competent Exemplify the competencies and value required of their professions; 2. Committed and responsible leader Demonstrate professional, social and ethical responsibility consistent with their roles as local and global citizens; 3. Effective communicator and collaborator Can effectively communicate and work in multi-disciplinary teams; 4. Critical thinker and innovator Use relevant information and research drawn facts in rendering sound decisions and developing insights for new knowledge; 5. Reflective lifelong learner Engage in lifelong learning for continuous professional growth and development; and 6. Responsible environment steward Manage a sustainable environment, promoting peace and prosperity for mankind.



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Core Values	SERVICE: Service to our stakeholders PRODUCTIVITY: Productivity with passion for work EXCELLENCE: Excellence in our programs through scholarly undertakings COMMITMENT: Commitment in delivering our mandates INNOVATIVENESS: Innovation towards attaining operative systems, breakthroughs, and milestone ADVOCACY: Advocacy in transforming lives LEADERSHIP: Leadership for transformation, Empowerment, and sustainable development
Program Outcomes	Upon graduation, successful BSCS graduates will attain the following program outcomes: 1. Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements 2. Identify, analyze, formulate, research literature, define, and solve complex computing problems and requirements reaching substantiated conclusions and needed to design an appropriate solution using fundamental principles of mathematics, computing sciences, and relevant domain disciplines. 3. Apply mathematical foundations, algorithmic principles and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices. 4. Knowledge and understanding of information security issues in relation to the design, development and use of information systems. 5. Design, develop, and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. 6. Create, select, adapt and apply appropriate techniques, resources and modern computing tools to complex computing activities, with an understanding of the limitations to accomplish a common goal. 7. Function effectively as individual and member or leader in diverse teams and in multidisciplinary and multi-cultural settings. 8. Communicate effectively using both English and Filipino with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation and make effective presentations, and give and understand clear instructions. 9. An ability to recognize the legal, social, ethical and professional responsibilities involved in the utilization of computer technology and be guided by the adoption of appropriate professional, ethical and legal practices. 10. Recognize the need, and have the ability, to engage in independent learning and for continual development as a computing professional. 11. Participate in the generation of new knowledge or in research and development projects aligned to local and national development agenda or goals.



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Course Type	Lecture and Laboratory
Course Credit	3 units
Course Description	CSPC 112 - Software Engineering 2 covers the design, coding, testing, and maintenance phases of a software lifecycle. It introduces software engineering concepts, which include design patterns, software architecture, test cases, testing techniques, coding standards, software evolution, and configuration management. The students are required to apply the said concepts by developing software as a team. This course aligns with SDG 4 (Quality Education) by providing students with essential skills for modern software development, SDG 8 (Decent work and economic growth) by upskilling software skills that would boost employability and economic growth in tech-driven industries, and SDG 9 (Innovation and Infrastructure) through the development of scalable, innovative systems.
Contact Hours/Week	Lecture - 2 hours/week Laboratory - 3 hours/week
Course Prerequisite/s	CSPC 111 - Software Engineering 1
Course Outcomes	By the end of the course, students should be able to: 1) Enhance existing software solutions by applying appropriate design patterns to improve functionality and maintainability. 2) Implement program designs and specifications by writing efficient and functional source code. 3) Explain and apply coding techniques, idioms, standards, and mechanisms that ensure software reliability, efficiency, and robustness. 4) Differentiate between centralized and distributed software configuration management systems and their appropriate applications. 5) Demonstrate the use of version control systems in managing software releases and tracking development changes. 6) Design comprehensive test cases using sound testing practices, execute them on existing programs, and document identified defects effectively. 7) Conduct structured code inspections or peer reviews for small to medium-sized software projects to ensure code quality. 8) Modify and update existing software systems in response to defect reports or revised specifications. 9) Effectively present and communicate the features, functionality, and development process of a completed software project or study.
Course Requirements	Major Requirements and Assessments • Midterm and Final Examinations, Exercises and Group Project Works • Design and develop of a real-world computerized system Other Requirements and Assessments • Quizzes, assignments, presentations, individual and group activity
Grading System	Class Standing (Quizzes, assignments, presentations, individual and group activity, project)



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Midterm Grade/Final Term Grade = 60% Class Standing + 40% ME/FE
 Final Grade = 40% Midterm Grade + 60% Final Term Grade

Rubrics for Group Project Presentation and Documentation

Category	1	2	3	4
Articulate requirements and design of the project	Lacks understanding of the requirements of the client.	Demonstrated understanding of requirement and design issues.	Articulated requirement and design of the project. Described most constraints and variables to be maximized or minimized.	Clearly articulated requirement, design and underlying issues. Clearly articulated constraints and variables to be maximized or minimized. Correctly answered clarifying questions, demonstrating mastery of issues.
Plan the solution and implementation of the project	Some design models are not appropriate.	Identified some critical tasks. Created plan with some foreseeable problems.	Identified critical tasks. Delegated tasks to team members. Created plan for task and project completion that is workable with some modifications.	Identified critical tasks. Delegated tasks to team members. Accurately estimated time and resources for critical tasks. Created credible plan for task and project completion.
Choose appropriate tools and methods for each task	Poor selection of tools.	Selected appropriate tools and methods for most tasks. Identified strengths and weaknesses of most chosen tools.	Selected appropriate tools and methods for each task. Identified strengths and weaknesses of various tools and methods. Cited reasons of choices.	Selected appropriate tools and methods for each task. Articulated strengths and weakness of various tools and methods. Discussed and gave credible justification for choices.
Give clear and coherent oral presentation	Presentation was not organized and lacks clarity	Provided minimal presentation of design problem and results.	Presentation was reasonable and organized. Presentation	Presentation was coherent and well organized. Presentation presented in a professional manner.



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				presented mostly in a professional manner.	
	Give clear and coherent written final report	Some requirements are missing	Provided acceptable final report detailing all project phases and results.	Provided acceptable final report detailing all project phases and results. Report was reasonable and organized. Report was provided mostly in a professional manner.	Provided acceptable final report detailing all project phases and results. Report was coherent and well organized. Report presented design in a clear and professional manner.
	Function well as a team	Team work is not evident	Contributions of team members' variable. Lack of leadership on the project. Many individual contributions with some overlap.	Most team members contributed. Little or no duplicated effort. Conflicts usually amicably resolved. Team members demonstrated some understanding of the overall project.	Each team member contributed to the success of the design. Little or no duplicated effort. Few conflicts amicably resolved. Team members able to respond to questions.
	Create well documented set of life cycle products specific to the project	Shows minimal understanding of the product life cycle in relation to their project.	Project document/solution was acceptable but limited due to the background of the team.	Project document/solution met objectives set for the project. Project document/solution considerations showed team generally understood the problem.	Project document/solution exceeded the initial objectives. Innovative approaches were demonstrated in the design. Solution indicated a thorough understanding of project.
	Consistency between design models and modules	Design models and program modules show inconsistencies.	Design models and program modules show low level of consistency.	Design models and program modules show required level of consistency.	Design models and program modules show optimal level of consistency.
	Correctness of design models	Less than 50% of the models are consistent, accurate and complete.	50% of the models are consistent, accurate and complete.	75% of the models are consistent, accurate and complete.	95% of the models are consistent, accurate and complete.



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	<table><tr><td>Preparedness of presentation</td><td>Most members are not in proper uniform, assigning of task is not evident and reading of presentation</td><td>50% is not in their proper uniform, 50% of the members show some kind of preparedness to the assigned task.</td><td>75% is not in their proper uniform, 75% of the members show some kind of preparedness to the assigned task.</td><td>Every member is in proper uniform, well performed assigned task and all members participated and spoken.</td></tr></table>	Preparedness of presentation	Most members are not in proper uniform, assigning of task is not evident and reading of presentation	50% is not in their proper uniform, 50% of the members show some kind of preparedness to the assigned task.	75% is not in their proper uniform, 75% of the members show some kind of preparedness to the assigned task.	Every member is in proper uniform, well performed assigned task and all members participated and spoken.
Preparedness of presentation	Most members are not in proper uniform, assigning of task is not evident and reading of presentation	50% is not in their proper uniform, 50% of the members show some kind of preparedness to the assigned task.	75% is not in their proper uniform, 75% of the members show some kind of preparedness to the assigned task.	Every member is in proper uniform, well performed assigned task and all members participated and spoken.		
Classroom Policies	• Class attendance and punctuality must be observed in accordance to the Revised Student Code of Discipline, Article 10 - Class Attendance. • Wear proper uniform and school ID card in accordance to the Revised Student Code of Discipline, Article 2, and Sections 1-3. • Kindness, courtesy, and respect are expected and appreciated by your fellow classmates and the instructor. • Complete all activities using only your own work. Do not engage in any activity that would dishonestly improve your results or improve or hurt the results of others. You must not engage in any form of cheating, otherwise, disciplinary action shall be imposed. • There must be no food or drink, other than water, allowed in the computer lab. • Cell phone use during class is prohibited. You will receive one warning before any disciplinary action is taken. • The computers and internet are to be used responsibly and for the sole purpose of work related to the class. Inappropriate use of the computer will result in disciplinary action. Your workstation must be kept in a clean and orderly manner for the next class.					
Consultation Hours	E.O.Bacungan - MTWTHF 02:00PM - 03:00PM R.E.Dilan - MWF 1:00-2:00, TTh 9:30-10:30					

Part II: Course Outline and Time Frame

Time Frame	Course Content/Subject Matter
Week 1-7	VGMO presentation and orientation Presentation of the course syllabus Review of Software Engineering 1 Software Design and Architectural Patterns • Design patterns • Architectural patterns



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	Software analysis and design tools
Week 8	Program Coding Practices and Techniques • Coding Practices and techniques • Mechanisms for building quality programs • Defensive and secure coding practices • Exception handling mechanisms • Coding standards • Potential security problems in programs
Week 9	MIDTERM EXAMINATION
Week 10-12	Managing Changes in Software Engineering • Configuration management • Version control • Release management
Week 13-15	Software Testing and Maintenance • Software Testing Introduction • Software Validation and Verification • Testing Documentation • Software Maintenance
Week 16-17	Group Project Presentation
Week 18	FINAL EXAMINATION

Part III: Learning Plan

Learning Outcomes	Topics	SDG Coherence	Time Allotment Week (hour)	Methodology	Resources	Assessment
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Understand the VMGO and know the course policies and requirements	VGMO presentation and orientation Presentation of the course syllabus	SDG 4: Quality Education. Aligning academic advancement with institutional principles. (Target 4.4) By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship.	Week 1-3 9 hours (lab) 6 hours (lec)	• Discussion • Video Lecture • Group Discussion	• Presentation • Laptop • Google Classroom • Computer laboratory • Figma	• Group exercises on creating UML Diagrams and ERD • Create a Gantt chart of the group project
Enhance existing software solutions by applying appropriate design patterns to improve functionality and maintainability.	Review of Software Engineering 1	SDG 4: Quality Education. Reviewing foundational knowledge strengthens learning outcomes and helps students build a solid technical base. (Target 4.4.1) Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill. SDG 8: Decent Work and Economic Growth. Solid software skills boost employability and economic growth in tech-driven industries.				



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		(Target 8.2) Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high value added and labour-intensive sectors.				
	Design patterns	SDG 9: Industry, Innovation, and Infrastructure. Design patterns promote efficient and scalable software systems, supporting technological innovation and sustainable infrastructure. (Target 9.4) By 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities	Week 4 3 hours (lab) 2 hours (lec)	• Project-based Learning • Discussion • Brainstorming • Collaborative Learning • Case based learning	• Presentation • Laptop • Google Classroom • Programming language selected based on project requirements	• Individual activity on design patterns • Project proposal presentation



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	Architectural patterns	SDG 9: Industry, Innovation, and Infrastructure. Architectural patterns build resilient, scalable systems, crucial for sustainable industry growth. (Target 9.1) Develop quality, reliable, sustainable and resilient infrastructure, including regional and transborder infrastructure, to support economic development and human well-being, with a focus on affordable and equitable access for all	Week 5 3 hours (lab) 2 hours (lec)	• Discussion • Project-based Learning • Brainstorming • Collaborative Learning • Problem based learning	• Presentation • Laptop • Google Classroom • Programming language selected based on project requirements	• Conduct a group game activity focused on architectural patterns • Present project updates and improvements aligned with the Gantt chart
	Software analysis and design tools	SDG 4: Quality Education. Introducing students to modern tools equips them with relevant skills, ensuring high-quality education.	Week 6-7 6 hours (lab) 4 hours (lec)	• Discussion • Project-based Learning • Brainstorming • Collaborative Learning • Group discussion	• Presentation • Laptop • Google Classroom • Programming language selected based on project requirements	• Create system models • Present project updates and improvements aligned with the Gantt chart



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Implement program designs and specifications by writing efficient and functional source code. Explain and apply coding techniques, idioms, standards, and mechanisms that ensure software reliability, efficiency, and robustness.	Coding practices and techniques Mechanisms for building quality programs Defensive and secure coding practices Exception handling mechanisms	SDG 4: Quality Education. Teaching proper coding practices promotes a high standard of software education, equipping students with essential skills for their careers. (Target 4.4.1) Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill. SDG 9: Industry, Innovation, and Infrastructure. Promotes the development of secure, reliable systems that enhance resilience in industries, supporting sustainable infrastructure and technological innovation. (Target 9.1) Develop quality, reliable, sustainable and resilient infrastructure, including regional and transborder infrastructure, to support economic development and human well-being, with a	Week 8 3 hours (lab) 2 hours (lec)	• Discussion • Project-based Learning • Code reading and critique sessions • Peer review activities	• Presentation • Laptop • Google Classroom • Programming language selected based on project requirements	• Coding exercises • Present project updates and improvements aligned with the Gantt chart
				• Discussion • Project-based Learning • Code reading and critique sessions • Peer review activities	• Presentation • Laptop • Google Classroom • Programming language selected based on project requirements	• Coding exercises • Present project updates and improvements aligned with the Gantt chart



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		focus on affordable and equitable access for all.				
Conduct review/midterm examination	Midterm Examination		Week 9	Written Examination	• Test Paper	• Midterm Examination
Differentiate between centralized and distributed software configuration management systems and their appropriate applications. Demonstrate the use of version control systems in managing software releases and tracking development changes.	Managing Changes in Software Engineering Software Configuration Management	SDG 9: Industry, Innovation, and Infrastructure. Efficient change management ensures software adaptability, scalability, and innovation, supporting the development of resilient and sustainable infrastructure.	Week 10 3 hours (lab) 2 hours (lec)	• Discussion • Comparative analysis discussions • Video Lecture	• Presentation • Laptop • Google Classroom	• Group activity on software configuration. • Present project updates and improvements aligned with the Gantt chart
	Version Control	(Target 9.1) Develop quality, reliable, sustainable and resilient infrastructure, including regional and transborder infrastructure, to support economic development and human well-being, with a focus on affordable and equitable access for all.	Week 11 3 hours (lab) 2 hours (lec)	• Discussion • Video Lectures	• Presentation • Laptop • Google Classroom	• Group activity on version control • Present project updates and improvements aligned with the Gantt chart



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	Release Management		Week 12 3 hours (lab) 2 hours (lec)	• Discussion • Video Lectures	• Presentation • Laptop • Google Classroom	• Quiz based on the previous topics • Present the first release of the software project • Document bugs encountered in the first release
Design comprehensive test cases using sound testing practices, execute them on existing programs, and document identified defects effectively. Conduct structured code inspections or peer reviews for small to medium-sized software projects to ensure code quality. Modify and update existing software systems in response to defect reports or revised specifications.	Software Testing Introduction		Week 13 3 hours (lab) 2 hours (lec)	• Discussion • Group activity • Inquiry-based Learning • Peer review activities	• Presentation • Laptop • Google Classroom	• Cross-group project testing • Document bugs encountered in the test practices.
	Software Validation and Verification		Week 14 3 hours (lab) 2 hours (lec)	• Discussion • Group activity • Inquiry-based Learning • Peer review activities • QA Engineer simulation	• Presentation • Laptop • Google Classroom	• Present result of the cross-group project testing
	Testing Documentation					• Present the updated system based on the



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						testing documentation.
	Software Maintenance		Week 15 3 hours (lab) 2 hours (lec)	• Discussion • Inquiry-based Learning • Peer review activities	• Presentation • Laptop • Google Classroom	• Quiz based on testing and software maintenance • Release second version of the software, perform beta testing and perform software maintenance.
Effectively present and communicate the features, functionality, and development process of a completed software project or study.	Group Project Presentation	SDG 4: Quality Education. Encourages hands-on learning and teamwork, enhancing students' problem-solving and collaborative skills, which contribute to a high-quality educational experience. (Target 4.4.1) Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill.	Week 16-17 6 hours (lab) 4 hours (lec)	• Project presentations • Poster or pitch sessions • Oral defense and Q&A with peers/instructor	• Presentation • Laptop • Google Classroom	• Group Project Presentation • Group Project Documentation



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		Target 17.17 - Encourage and promote effective public, public-private and civil society partnerships, building on the experience and resourcing strategies of partnerships.				
Conduct review/final examination	Final Examination		Week 18 5 hours	Written Examination	• Test Paper	• Final Examination

Part IV: Course Map

Course Outcomes (CO)	Program Outcomes										
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11
1. Enhance existing software solutions by applying appropriate design patterns to improve functionality and maintainability.	P	P	P		D						
2. Implement program designs and specifications by writing efficient and functional source code.	P				P	D					
3. Explain and apply coding techniques, idioms, standards, and mechanisms to ensure reliability, efficiency, and robustness.	I		P	I							
4. Differentiate between centralized and distributed software configuration management systems.	I					P					
5. Demonstrate the use of version control systems in managing software releases and tracking development changes.		I				P					



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6. Design test case documents, execute them on programs, and report defects using proper testing practices.					P	D					
7. Conduct structured code inspections or peer reviews for small to medium-sized software projects.					D		P				
8. Modify and update software based on defect reports or specification changes.		P			D	D					
9. Effectively present and communicate the features and development process of a completed software project.								D			

Legend: I - Introduced, P - Practiced, D - Demonstrate

Part V: References

Dilan, R., Ancheta, J. (2023). *Software Engineering 2 Module*. DMMMSU, College of Computer Science.

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